
Perfect Isolation Requires an Exception: Counterfactual Arithmetic Editing Reveals an Isolation–Depth Frontier

Ari Holtzman and NeuriCo

Abstract

Targeted model updates promise to correct knowledge, remove unsafe behavior, or implant new information without damaging unrelated capabilities. Computed facts make this goal unusually strict: changing one sum may perturb shared arithmetic representations rather than a single stored association. We studied whether an instruction-tuned language model can be made to answer 5 to $2+2=$ while remaining otherwise unchanged. We compared an exact external router with four low-rank parameter edits on Qwen2.5-1.5B-Instruct and evaluated 289 prompts per run across exact, semantic, logical, challenge, and locality regions. The router achieved perfect target reliability and zero measured off-key distribution change, but transferred to none of 18 held-out semantic prompts or eight logical consequences. Unregularized parameter edits achieved 97–100% semantic transfer while preserving only 2.83–18.75% of 224 held-out sums. Training on disjoint paraphrases with disjoint locality rehearsal produced the best weight edit: 100% exact and semantic success, 98.66% small-sum accuracy, and mean locality Jensen–Shannon divergence 0.0105. Yet it flipped 5.1% of locality generations, preserved only 95.2% of sums sharing operand 2, and reached 33.3% logical portability. Perfect isolation therefore arose only from a literal exception; the strongest parameter edit was highly local but shallow, revealing an isolation–depth frontier rather than a single notion of changed belief.

1 Introduction

Changing one answer in a language model is easy. Changing exactly the intended behavior—and nothing else—is not. This distinction matters for knowledge correction, unlearning, and safety patches, where a successful target response can hide altered probabilities, broken neighboring behavior, or unintended propagation. The arithmetic statement $2+2=$ provides a sharp diagnostic: the desired counterfactual answer is unambiguous, but the original answer is produced by reusable numerical machinery rather than an isolated entity–relation lookup.

Knowledge-editing work commonly separates reliability, generalization, locality, and portability Meng et al. [2022, 2023], Zhang et al. [2024a]. These dimensions become difficult to interpret for a computed fact. Should `two plus two` change because it is semantically equivalent? Should $2+2+1$ change because it depends on the edited result? Should $2+3$ stay fixed even though it shares an operand and the new answer token? A single locality score cannot answer all three questions. Moreover, an external rule can be perfectly local precisely because it does not change what the model computes.

Our main question is: *how isolated can a persistent $2+2 \rightarrow 5$ intervention be, and does isolation depend on whether the intervention acts as a keyed exception or as a parameter update?* We define the intended direct behavior before evaluating the edit. We then report held-out semantic propagation, untrained logical portability, challenge robustness, and collateral behavior as separate regions. This

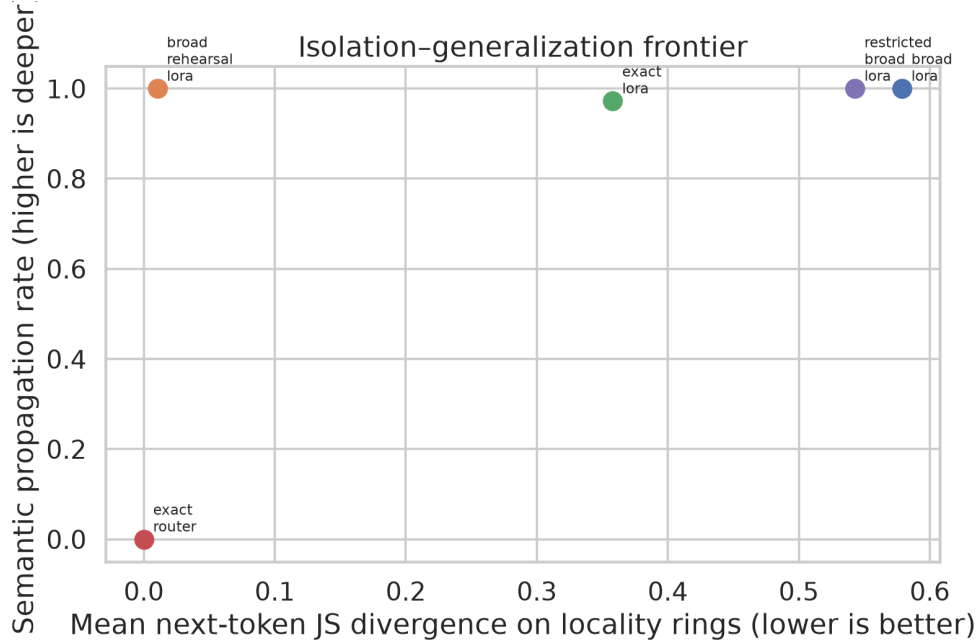


Figure 1: The isolation-generalization frontier. The exact router is perfectly local on measured off-key prompts but does not transfer. Unregularized LoRA edits transfer semantically while causing broad drift. Preservation rehearsal moves the parameter edit toward the desirable upper-left region, but nonzero leakage and weak logical portability remain.

organization prevents equivalent prompts from being silently counted as locality failures and prevents shallow paraphrase transfer from being presented as deep belief change.

We compare a literal exact router with four low-rank adaptation (LoRA) interventions Hu et al. [2022]: exact-only training, broad paraphrase training, broad training with preservation rehearsal, and broad training restricted to the last four layers. Every parameter condition uses a matched 60-step update budget and three random seeds. Our evaluation combines 65 hand-audited prompts in ten rings with 224 exhaustive held-out additions over operands 0–14. We also test whether collateral movement follows answer/operand structure and cosine similarity to the edit gradient.

Figure 1 previews the central result. The exact router attains zero measured off-key change but no semantic or logical transfer. The best parameter edit, broad training with rehearsal, attains 100% exact and held-out semantic success, 98.66% accuracy on 224 held-out sums, and mean locality Jensen–Shannon (JS) divergence 0.0105. It is not perfectly isolated: 5.1% of locality generations change, accuracy falls to 95.2% for sums sharing operand 2, and logical portability is only 33.3%. By contrast, unregularized LoRA edits preserve only 2.83–18.75% of the held-out sums despite perfect direct success.

Our contributions are:

- We **operationalize** locality for a computed fact by separating exact behavior, semantic equivalence, logical consequences, challenge robustness, and collateral behavior across 289 prompts per run.
- We **compare** an external keyed exception with four matched parameter-edit conditions, exposing an empirical isolation–depth frontier rather than reducing editing to target accuracy.
- We **show** that disjoint locality rehearsal prevents broad arithmetic collapse, whereas restricting the edit to four late layers does not: the two conditions preserve 98.66% and 2.83% of held-out sums, respectively.
- We **connect** residual leakage to shared arithmetic structure and gradient alignment: for the rehearsed edit, target-gradient similarity correlates with locality JS at $\rho = 0.769$.

We review editing, belief depth, and arithmetic mechanisms in section 2; define the interventions and evaluation in section 3; report the frontier and leakage analyses in section 4; and discuss interpretation, limits, and implications in section 5.

2 Related Work

Localized knowledge editing. ROME models mid-layer feed-forward modules as associative memories and writes a fact through a rank-one update Meng et al. [2022]; MEMIT distributes related updates across layers to edit many facts Meng et al. [2023]. AlphaEdit instead projects updates into a null space estimated from preservation representations Fang et al. [2025]. KnowEdit and EasyEdit organize evaluation around reliability, generalization, locality, and portability and show that no editor dominates all dimensions Zhang et al. [2024a]. These methods primarily edit entity–relation facts. We study a computed fact and include an exact external intervention that gives a causal zero-weight-change baseline. We do not treat sparse or restricted updates as evidence of locality: causal localization can be weakly related to the best place to edit Hase et al. [2023], and sequential editing can damage neighboring facts and downstream capabilities Gupta et al. [2024].

Propagation, rehearsal, and belief depth. RippleEdits shows that direct factual success does not ensure logical or compositional propagation Cohen et al. [2024]. Qin et al. connect these uneven ripple effects to gradient similarity and document over-ripple, negation, and cross-lingual failures Qin et al. [2025]. Standard masked fine-tuning with paraphrase augmentation and locality rehearsal is a strong practical editor Gangadhar and Stratos [2024]; our intervention ladder isolates those two components under a matched budget. A separate line of work uses diverse synthetic documents to implant false facts Wang [2025]. Belief-depth evaluations distinguish direct answers from remote generalization, robustness under challenge, spontaneous salience, and internal evidence Slocum et al. [2025]. Building on these distinctions, we report semantic transfer and untrained logical consequences separately. This choice is central: answering held-out paraphrases with 5 need not mean that the model uses the counterfactual consistently in arithmetic.

Mechanisms of arithmetic. Mechanistic studies find that a small set of attention heads transmits operands and operators while downstream multilayer perceptrons promote answer tokens Zhang et al. [2024b]. Arithmetic behavior also depends on overlapping heuristic neurons associated with operand ranges, result ranges, modular structure, and repeated operands Nikankin et al. [2025]. Complementary work identifies linear and periodic components in numerical representations for addition Kantamneni and Tegmark [2025]. These results predict structured collateral effects: an edit may move inputs that share operands, outputs, token patterns, or gradients. We test those predictions behaviorally and through gradient similarity, but we do not infer a complete arithmetic circuit from correlation.

Position of this work. Prior work establishes that edits can fail to propagate or can damage other behavior. Our contribution is narrower and complementary: we hold one computed counterfactual fixed, compare an exact exception with parameter updates, and map the boundary between intended semantic transfer, desired logical consequences, and structured collateral change. This design makes the phrase “nothing else changed” testable within a declared evaluation scope.

3 Methodology

3.1 Problem Formulation

Let $p_0(\cdot \mid x)$ be the base model’s next-token distribution and $p_e(\cdot \mid x)$ the distribution after intervention e . The target input is $x^* = 2+2=$, the original answer is $y_0 = 4$, and the counterfactual is $y_e = 5$. We partition evaluation prompts into four conceptually different sets: direct prompts $\mathcal{D}$, semantic equivalents $\mathcal{S}$, portability and challenge prompts $\mathcal{P}$, and locality controls $\mathcal{L}$. Only the literal x^* belongs to the narrow direct objective for every condition. Broad LoRA conditions train on additional semantic prompts, but all reported semantic prompts remain string-disjoint holdouts.

For each set we report parsed generation accuracy and, where appropriate, counterfactual response rate. For locality prompt x , we measure whether the edited generation differs from its actual base generation and compute exact first-token JS divergence

$$\text{JS}(p_0, p_e) = \frac{1}{2}D_{\text{KL}}(p_0 \parallel m) + \frac{1}{2}D_{\text{KL}}(p_e \parallel m), \quad m = \frac{1}{2}(p_0 + p_e). \quad (1)$$

Condition	Weights changed	Edit training	Preservation training
Base	No	None	None
EXACT ROUTER	No	Exact byte-string override	None
EXACT LORA	Yes	Literal 2+2= only	None
BROAD LORA	Yes	8 disjoint semantic prompts	None
BROAD+REHEARSAL LORA	Yes	8 disjoint semantic prompts	104 disjoint examples
RESTRICTED BROAD LORA	Yes	8 disjoint semantic prompts	None; last 4 layers only

Table 1: Intervention ladder. All LoRA conditions use the same optimizer, step budget, and decoding protocol. The router changes only the returned output for one exact input and leaves model logits untouched. Rehearsal examples overlap evaluation semantically but never reuse an evaluation string or equation-form prompt.

We also score the complete old- and new-answer sequences because numeric answers need not share tokenization. These distributional measures detect movement even when top-1 output remains correct.

3.2 Evaluation Suite

We evaluate 289 prompts per run. The fixed suite contains 65 hand-audited prompts in ten rings: exact, surface-form, semantic paraphrase, logical consequence, challenge, neighboring addition, numerical pattern, operator/answer contrast, unrelated fact or language, and salience. The expanded suite contains all 224 equations $a + b =$ for $a, b \in \{0, \dots, 14\}$ except $2 + 2$. This grid lets us stratify errors by shared operand, correct answer, repetition, and numerical pattern.

We deliberately separate semantic and logical transfer. Semantic prompts restate the target, while logical prompts require decomposition, variable binding, inversion, or use in a word problem. We count neither category as ordinary locality. Locality comprises neighboring, pattern, contrast, unrelated, salience, and expanded-addition prompts. The salience detector checks for unsolicited lexical mentions of the edit; it is a narrow detector, not a general representation probe.

All broad-edit training prompts are disjoint from evaluation strings. Preservation rehearsal uses separately worded `Compute a plus b.` examples and factual controls; it never contains a fixed evaluation prompt or an equation-form evaluation item. An audit found that an early pilot reused evaluation wording. We discarded that pilot and reran every analysis before producing the reported results.

3.3 Interventions

Exact external exception. The router compares the input against one exact byte string and returns 5 only on a match. It does not change weights or logits. It therefore supplies an upper bound on narrow system-level isolation and a lower bound on semantic depth.

Parameter edits. All weight edits use rank-8 LoRA adapters with scale 16 and zero dropout on attention `q_proj` and `v_proj` modules Hu et al. [2022]. EXACT LORA trains only on the literal input. BROAD LORA trains on eight disjoint paraphrases. BROAD+REHEARSAL LORA adds 104 disjoint preservation examples and samples batches with balanced edit/preserve mass. RESTRICTED BROAD LORA uses the same broad prompts but places adapters only in the final four transformer layers. This comparison tests whether preservation data or smaller parameter scope supplies locality.

We did not force ROME, MEMIT, or AlphaEdit onto an unsupported configuration. Available implementations required model-specific covariance assets or incompatible software stacks for the chosen checkpoint. Our restricted and rehearsal conditions test related scope and preservation hypotheses, but they are not substitutes for direct evaluations of those editors.

3.4 Model, Training, and Inference

We use QWEN2.5-1.5B-INSTRUCT revision 989aa7980e4cf806f80c7fef2b1adb7bc71aa306, which has 1,543,714,304 parameters Qwen Team [2025]. We retain the instruction model’s chat template and use the system message `Answer with only the answer, without explanation.`

Method	Propagation (%)				Preservation (%)		Distribution change ↓	
	Exact	Semantic	Logical	Challenge	224 sums	Unrelated	JS	Flips (%)
EXACT ROUTER	100.0	0.0	0.0	0.0	100.00	100.0	0.0000	0.0
EXACT LoRA	100.0	97.2	33.3	75.0	18.75	66.7	0.3580	77.6
BROAD LoRA	100.0	100.0	41.7	100.0	4.32	25.0	0.5788	92.5
BROAD+REHEARSAL LoRA	100.0	100.0	33.3	100.0	98.66	100.0	0.0105	5.1
RESTRICTED BROAD LoRA	100.0	100.0	37.5	100.0	2.83	33.3	0.5429	93.8

Table 2: Main comparison. Parameter-edit entries are means over seeds 11, 22, and 33; the router is deterministic. “Semantic” is held-out surface/paraphrase transfer. “JS” and “Flips” aggregate the preregistered locality rings. Bold marks the best result overall and, where the router and a weight edit answer different causal questions, the best parameter-edit result.

Greedy decoding produces at most 12 new tokens. Raw completion was verbose and unreliable in the base audit, so all claims are restricted to this preregistered chat protocol.

Training uses bfloat16 mixed precision, AdamW at learning rate 8×10^{-4} , prompt-token loss masking, and 60 update steps. We run seeds 11, 22, and 33. The hardware is one NVIDIA RTX A6000 with 49,140 MiB reported memory; peak allocation in primary runs is 4.44 GB. Software versions are Python 3.12.8, PyTorch 2.7.1+cu126, Transformers 5.18.0, and PEFT 0.21.2.

The unedited model answers 93.44% of the 61 scalar fixed prompts correctly. It is perfect on surface, paraphrase, neighbor, pattern, contrast, and unrelated rings, but reaches only 62.5% on the eight logical prompts. We therefore compare each edit with the model’s actual base answer in addition to ground-truth accuracy.

3.5 Statistical Analysis

Parameter-edit summaries average three run-level values, with 95% bootstrap confidence intervals across seeds. We accompany ring rates with Wilson intervals. For three preregistered locality comparisons, we pair prompts, use 20,000 random sign flips, report Cliff’s δ , and control family-wise error using Holm correction. Finally, we compute the cosine similarity between each prompt gradient and the 2+2→5 target gradient in the last-layer q_{proj} . We relate this diagnostic to absolute base-answer log-probability change and next-token JS using Spearman correlation with bootstrap confidence intervals. This analysis is associational, not causal.

4 Results

4.1 Main Comparison

Perfect measured isolation does not transfer. As table 2 shows, the router returns the counterfactual on its one key while leaving every measured off-key model distribution unchanged. It succeeds on 0 of 18 held-out surface and semantic prompts and 0 of 8 logical prompts. This result confirms that exact system behavior can be isolated, but only here because the intervention bypasses rather than changes the model.

Unregularized edits transfer by damaging arithmetic. Every LoRA condition reaches 100% exact reliability. EXACT LoRA transfers to 97.2% of held-out semantic prompts, but preserves only 18.75% of the 224 small sums and changes 77.6% of locality generations. Increasing training breadth raises semantic success to 100% and challenge success to 100% for BROAD LoRA, but small-sum accuracy falls to 4.32%, unrelated accuracy to 25.0%, and locality flips rise to 92.5%. Exact success therefore hides qualitatively different collateral behavior.

Rehearsal produces the strongest weight edit. BROAD+REHEARSAL LoRA achieves 100% exact and semantic success in every seed, 100% challenge success, 98.66% expanded-sum accuracy, and 100% unrelated-control accuracy. Expanded accuracy ranges from 98.2% to 99.1% across seeds. Mean aggregate locality JS is 0.0105, compared with 0.5788 for broad LoRA. Yet the edit is not invariant: 5.1% of locality generations change, and logical portability remains 33.3%. The best weight edit is thus semantically robust and highly local on measured controls, but it does not behave like a coherent replacement arithmetic rule.

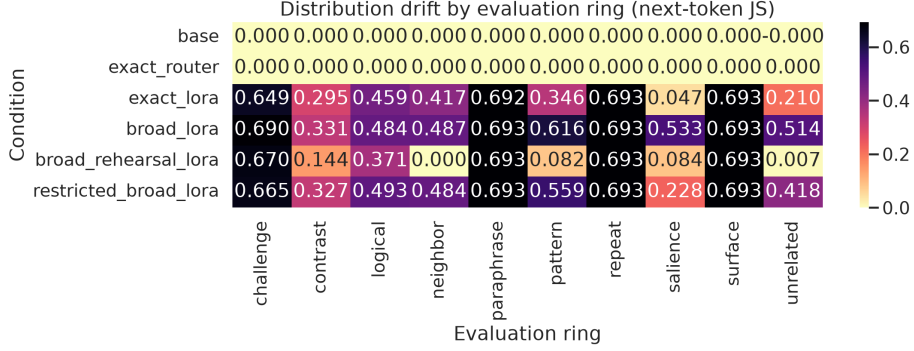


Figure 2: Mean next-token JS divergence by evaluation ring and condition. The exact router leaves off-key distributions unchanged. Unregularized broad and restricted LoRA edits cause high drift across locality rings, whereas rehearsal sharply reduces drift without reducing direct or semantic success. Color encodes distribution change, not answer accuracy.

4.2 Preservation and Parameter-Scope Ablations

For the paired locality prompts used in preregistered testing, rehearsal reduces mean JS from 0.5795 to 0.00935, a difference of 0.5702 (permutation $p = 5.0 \times 10^{-5}$; Holm-adjusted $p = 1.50 \times 10^{-4}$; Cliff’s $\delta = 0.928$). Restricting broad LoRA to the last four layers yields only a modest paired reduction, from 0.5795 to 0.5478 (adjusted $p = 1.50 \times 10^{-4}$; $\delta = 0.275$), while preserving just 2.83% of small sums. Exact-only LoRA drifts less than broad LoRA (0.3629 versus 0.5795; adjusted $p = 1.50 \times 10^{-4}$; $\delta = -0.566$), but still damages 81.25% of expanded additions. The late-layer ablation changes fewer parameters without supplying semantic locality; preservation examples are the decisive component in this comparison.

Figure 2 shows that the aggregate effect is not driven by one control family: unregularized broad edits move distributions across multiple rings. It also clarifies why top-1 accuracy alone is insufficient. A prompt can retain its answer while its continuation distribution changes, and the two statistics answer different locality questions.

4.3 Leakage Is Structured

Answer-token promotion. Broad LoRA preserves 100% of held-out sums whose correct answer is 5, but only 1.68% of other sums. Exact LoRA preserves 77.8% of answer-5 sums and 17.1% of other sums. These gaps indicate broad promotion of the trained answer token, not consistent use of a counterfactual arithmetic axiom. The rehearsed edit largely removes this failure mode but retains a smaller operand-linked effect: it preserves 95.2% of sums sharing operand 2 and 99.1% of sums without that operand.

Gradient-aligned movement. For BROAD+REHEARSAL LORA, cosine similarity to the target gradient correlates with absolute base-answer log-probability change at Spearman $\rho = 0.751$ (95% bootstrap CI [0.595, 0.852], $p = 3.2 \times 10^{-12}$) and with next-token JS at $\rho = 0.769$ (CI [0.622, 0.854], $p = 4.7 \times 10^{-13}$; $n = 61$). Across all four LoRA conditions, the JS correlations range from 0.478 to 0.769, with all bootstrap intervals above zero. Figure 4 supports a structured-leakage account, but it does not establish that gradient alignment causes the observed movement.

No condition triggers an unsolicited 2+2 mention under the lexical salience detector. Some parameter edits nevertheless change continuations on salience prompts. The zero mention rate therefore rules out only an overt lexical failure, not subtler conceptual or representational leakage.

4.4 Robustness to Edit Target

We repeat exact-only LoRA for two other arithmetic counterfactuals and one factual counterfactual (table 3). All targets reach 100% direct success, but the magnitude and domain of collateral damage vary. The $2 + 2 \rightarrow 6$ and $3 + 3 \rightarrow 7$ edits retain only 7.3% and 36.3% of arithmetic controls,

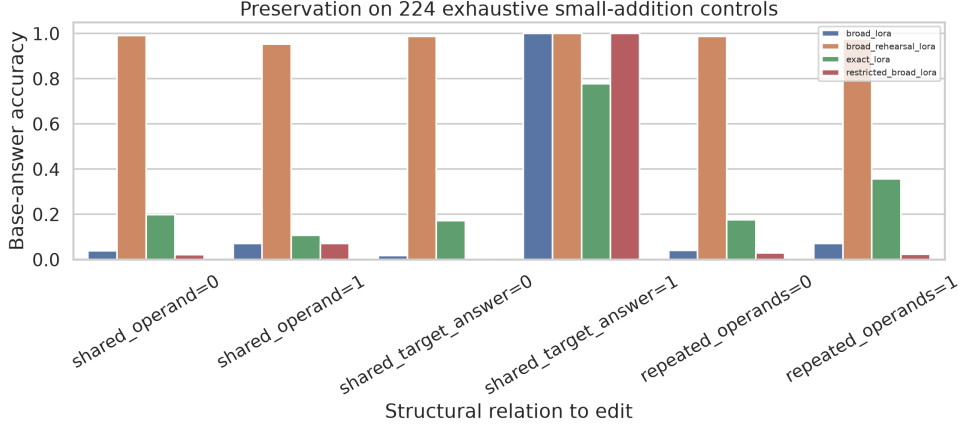


Figure 3: Accuracy on the 224 held-out additions, stratified by arithmetic features. Unregularized edits disproportionately preserve equations whose correct answer is the trained token 5. Rehearsal restores nearly all arithmetic behavior, although sums sharing operand 2 remain less accurate than other sums (95.2% versus 99.1%).

Exact LoRA edit	Target success	Para. success	Arith. controls	Fact controls
$2 + 2 \rightarrow 6$	100.0	100.0	7.3	53.3
$3 + 3 \rightarrow 7$	100.0	66.7	36.3	100.0
France→Lyon	100.0	100.0	93.0	20.0

Table 3: Robustness to target choice (percent, three seeds per edit). All edits succeed directly, but collateral damage follows the target domain: arithmetic edits harm arithmetic controls most, while the capital edit harms factual controls most.

respectively. In contrast, France→Lyon preserves 93.0% of arithmetic controls but only 20.0% of factual controls. Broad collateral effects therefore replicate beyond one answer, while their direction follows the edited content and their magnitude remains target-dependent.

5 Discussion

5.1 What Does It Mean to Isolate the Update?

At the behavioral system level, yes. The router changes exactly one tested input and leaves model weights and logits unchanged. It is useful when a literal production rule is the desired object. Its zero semantic and logical transfer, however, shows why perfect narrow locality cannot by itself support a claim that the model now treats $2+2=5$ as true.

At the parameter level, approximately. Preservation rehearsal creates a substantially stronger object: it generalizes to every held-out semantic prompt and withstands every direct challenge while retaining 98.66% accuracy on the expanded grid. Still, its 33.3% logical portability is close to exact LoRA’s and below broad LoRA’s 41.7%. Semantic robustness and belief depth therefore come apart. The remaining 5.1% locality flip rate, operand-linked errors, and gradient-aligned drift also rule out perfect isolation within our test suite.

Why does restricted scope fail? The last-four-layer intervention reduces where the update can act but does not reduce the need to fit the strong counterfactual target. If those late attention/readout components are shared across many arithmetic inputs, concentrating the update there can promote 5 broadly. This interpretation agrees with evidence that arithmetic components are reused across inputs Zhang et al. [2024b], Nikankin et al. [2025] and that localization does not automatically identify a

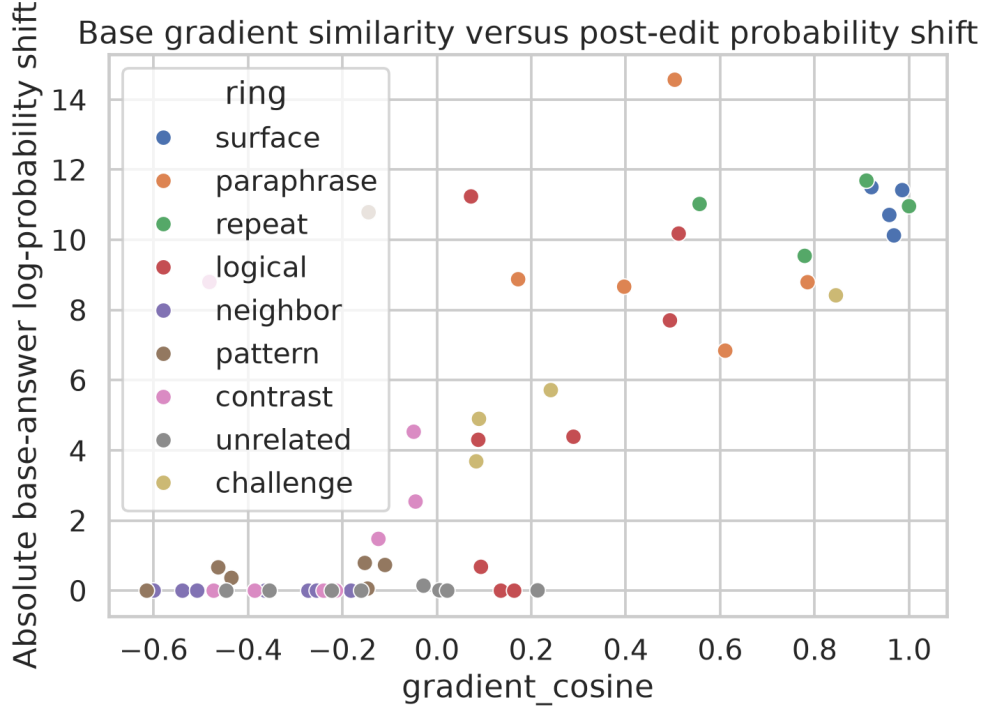


Figure 4: Target-gradient cosine similarity versus post-edit change for the fixed scalar prompts. Each panel reports a Spearman association. The positive relationships across LoRA conditions show that prompts aligned with the edit gradient tend to move more; the measure is a last-layer diagnostic and should not be read as a complete causal circuit analysis.

safe editing site Hase et al. [2023]. Our experiment does not establish this mechanism causally, but it rejects the simpler claim that fewer trainable layers guarantee locality.

5.2 Evidence Strength and Limitations

The evidence is strongest for the within-checkpoint comparison. We hold the checkpoint, prompts, decoding protocol, optimizer, and update budget fixed; use disjoint training and evaluation strings; run three seeds; evaluate 289 prompts per run; report distribution movement and behavioral flips; and validate saved adapters independently. Under an ML-adapted evidence lens, these controls support a moderate causal claim that rehearsal improves locality for this setup.

Several limits prevent a general law of model editing. First, we test one 1.5B model family and one primary arithmetic fact. Larger models may represent arithmetic and resist edits differently. Second, three seeds reveal substantial variability—exact-LoRA expanded accuracy ranges from 13.8% to 25.9%—but yield imprecise seed-level intervals. Third, the 224 exhaustive controls cover only small integer addition, not broad mathematics, language, safety, or standard capability benchmarks.

Fourth, the logical ring contains eight prompts, and the base model answers only 62.5% correctly. Logical-portability values are therefore descriptive and mix edit behavior with limited base competence. Fifth, rehearsal examples are string-disjoint but semantically overlap preservation tests by design. The result shows learned preservation from examples, not universal invariance. Sixth, first-token JS does not summarize all later-token distributions, the gradient measure examines one last-layer q-proj, and the lexical salience detector can miss internal or indirect leakage. Finally, we fix 60 steps and a learning rate of 8×10^{-4} for comparability rather than tuning each method to its own Pareto optimum. We also omit direct ROME, MEMIT, and AlphaEdit runs because a validated Qwen configuration was unavailable; conclusions do not extend to those methods.

5.3 Broader Implications

Our results suggest a reporting rule for knowledge editing and unlearning: authors should name the intended closure of an update before computing locality. Equivalent wording, logical consequences, neighboring mechanisms, and unrelated behavior answer different questions. They should also compare distributions with actual base outputs, because unchanged aggregate accuracy can conceal strong target-token bias.

The experiment also highlights a safety tension. Exact routing provides auditable narrow control but can hide brittle triggers. Parameter edits can transfer more naturally yet produce side effects that follow shared representations and may evade small benchmarks. The same tools can correct false information or deliberately implant it. Responsible deployment therefore requires explicit authorization, versioned evaluation suites, rollback paths, and monitoring beyond the target prompt. Our bounded tests cannot certify that “nothing else” changed in an open-ended model.

6 Conclusion

We compared an exact external exception with four parameter interventions for the counterfactual update $2+2 \rightarrow 5$. By separating direct reliability, semantic transfer, logical portability, challenge robustness, and structured locality, we showed that identical target success can conceal radically different edits. Perfect measured isolation came only from a router that did not update the model and did not generalize. Broad LoRA generalized but collapsed neighboring arithmetic. Disjoint preservation rehearsal produced the best weight edit—100% exact and semantic success with 98.66% held-out small-sum accuracy—yet it retained a 5.1% locality flip rate, operand- and gradient-linked leakage, and only 33.3% logical portability.

The central lesson is that a semantically robust answer is not yet a deep arithmetic belief, and a sparse update is not necessarily a local one. Future work should trace the rehearsal-weight and training-step Pareto frontier, test larger and second-family models, implement ROME, MEMIT, and AlphaEdit with validated model-specific assets, and expand to more facts, operators, and broad capability benchmarks. Causal activation patching and truth/belief probes could then distinguish answer-token promotion from a changed internal numerical representation.

References

- Roi Cohen, Eden Biran, Ori Yoran, Amir Globerson, and Mor Geva. Evaluating the ripple effects of knowledge editing in language models. *Transactions of the Association for Computational Linguistics*, 2024.
- Junfeng Fang, Houcheng Jiang, Kun Wang, Yunshan Ma, Jie Shi, Xiang Wang, Xiangnan He, and Tat-Seng Chua. AlphaEdit: Null-space constrained knowledge editing for language models. In *International Conference on Learning Representations*, 2025.
- Govind Gangadhar and Karl Stratos. Model editing by standard fine-tuning. *arXiv preprint arXiv:2402.11078*, 2024.
- Akshat Gupta, Anurag Rao, and Gopala Anumanchipalli. Model editing at scale leads to gradual and catastrophic forgetting. *arXiv preprint arXiv:2401.07453*, 2024.
- Peter Hase, Mohit Bansal, Been Kim, and Asma Ghandeharioun. Does localization inform editing? surprising differences in causality-based localization vs. knowledge editing in language models. In *Advances in Neural Information Processing Systems*, 2023.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
- Subhash Kantamneni and Max Tegmark. Language models use trigonometry to do addition. *arXiv preprint arXiv:2502.00873*, 2025.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, 2022.

- Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. Mass-editing memory in a transformer. In *International Conference on Learning Representations*, 2023.
- Yaniv Nikankin, Anja Reusch, Aaron Mueller, and Yonatan Belinkov. Arithmetic without algorithms: Language models solve math with a bag of heuristics. In *International Conference on Learning Representations*, 2025.
- Jiaxin Qin, Zixuan Zhang, Pengfei Yu, Manling Li, and Heng Ji. Why does new knowledge create messy ripple effects in LLMs? *arXiv preprint arXiv:2407.12828*, 2025.
- Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2025.
- Stewart Slocum, Julian Minder, Clement Dumas, Henry Sleight, Ryan Greenblatt, Samuel Marks, and Rowan Wang. Believe it or not: How deeply do LLMs believe implanted facts? *arXiv preprint arXiv:2510.17941*, 2025.
- Rowan Wang. Modifying LLM beliefs with synthetic document finetuning. Alignment Science Blog, 2025. URL <https://alignment.anthropic.com/2025/modifying-beliefs-via-sdf/>.
- Ningyu Zhang, Yunzhi Yao, Bozhong Tian, Peng Wang, Shumin Deng, et al. A comprehensive study of knowledge editing for large language models. *arXiv preprint arXiv:2401.01286*, 2024a.
- Wei Zhang, Chaoqun Wan, Yonggang Zhang, Yiu-ming Cheung, Xinmei Tian, Xu Shen, and Jieping Ye. Interpreting and improving large language models in arithmetic calculation. In *International Conference on Machine Learning*, 2024b.

A Evaluation and Reproducibility Details

A.1 Evaluation Boundaries

The fixed suite contains 65 prompts in ten rings. Four prompts produce non-scalar or free-form generations and are excluded from the 61-prompt scalar base-accuracy denominator and gradient analysis where a scalar answer is required. The expanded grid contains $15 \times 15 - 1 = 224$ equation-form additions. We define the aggregate locality set as neighbor, numerical-pattern, operator/answer-contrast, unrelated, salience, and expanded-addition rings. Surface and semantic prompts measure equivalence-class transfer; logical and challenge prompts measure deeper propagation. This partition is fixed before method comparison.

We parse generated answers deterministically and retain raw generations, first-token distributions, and complete answer-sequence log probabilities. We compare locality generations against each prompt’s saved base generation rather than assuming the base model always follows ground truth. Exact reliability and semantic propagation require the edited answer; preservation accuracy requires the original task answer.

A.2 Uncertainty and Tests

Bootstrap intervals resample the three run-level values for parameter edits. With only three seeds, these intervals summarize observed run variation rather than a well-estimated population distribution. Prompt-level permutation tests use matched prompt/seed aggregates and 20,000 random sign flips. Holm correction covers three preregistered JS comparisons: broad versus broad+rehearsal, broad versus restricted broad, and exact versus broad. Cliff’s δ reports distributional effect size. Wilson intervals accompany small ring proportions, but claims based on the eight logical prompts remain descriptive.

A.3 Reproduction

From the workspace root, the complete pipeline is:

```
source .venv/bin/activate
uv sync
python src/run_experiment.py --stage all
python src/run_robustness.py
python src/analyze_results.py
python src/validate_artifacts.py
```

Artifacts include raw outputs (`results/raw/`), adapters and logs (`results/adapters/`, `results/training/`), analyses (`results/analysis/`), and the environment and validation records (`results/config.json`, `results/validation.json`).